

UniVerse

A Campus Companion

Start Here - Defense Overview

Read this first. The whole app on two pages, then one PDF per feature.

Leading University, Sylhet - Dept. of CSE

Course CSE-3240 (Project I) - Team Sherlocked

Flutter (Dart) - Supabase - Firebase FCM - OR-Tools CP-SAT

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App: UniVerse - A Campus Companion (Android, Flutter)

1. The 30-second pitch

UniVerse is a role-aware campus companion for students, teachers and admins at Leading University. It unifies routines, resources, class cancellations and notifications in one app - and its differentiator is an **automatic, clash-free department timetable generator** powered by an OR-Tools CP-SAT constraint solver.

2. The one architecture rule (every feature follows it)

Every feature is a vertical slice with the SAME four layers. If you remember one thing for the defense, remember this chain:

```
Screen  -> Controller  -> Service  -> Supabase / Engine
  (UI)      (state, logic)  (data calls)  (backend)
```

layer	responsibility
Screen	Pure UI. Reads the controller, renders widgets. NEVER touches Supabase.
Controller	A ChangeNotifier holding state + logic. Calls services, then notifyListeners().
Service	The ONLY layer that talks to Supabase / HTTP. Returns typed models.
Model	Typed view of a DB row (e.g. RoutineEntry, Profile). Pure Dart.

State management is plain **ChangeNotifier + ListenableBuilder** - no Riverpod/Bloc. The UI rebuilds when a controller calls notifyListeners().

3. Tech stack

area	what we use
Frontend	Flutter (Dart 3.9), GoRouter, google_fonts (Inter), phosphor icons.
Backend	Supabase - Postgres + Auth + Storage + Realtime, secured with Row Level Security (RLS).
Push	Firebase Cloud Messaging (FCM) via a send-push Edge Function.
Engine	FastAPI + Google OR-Tools CP-SAT (Python), deployed on Render.
Auth	Google OAuth (PKCE) + email/password; admin whitelist gate.

4. Who owns what (so you answer your own slice)

member	owns
Fahmid Alam	Architecture, shared infra, auth, admin + timetable engine, theming, routing.
Swadheen Islam (Robi)	Routine, resources, dashboards, teacher screens, Find Teacher, Rooms.
Shahriar Rashid (Ratul)	Notifications, push, profile, QA, seeding.

5. The document set (study your slice, skim the rest)

PDF	covers
01 Routing	One GoRouter + a redirect guard; tabs vs pushed screens.
02 Theming	Design tokens (colors/text/spacing) wired via AppTheme.
03 Authentication	OAuth + email, the AuthStatus machine, the whitelist gate.

04 Routine

One table, two read paths; the time/status helpers.

05 Student Dashboard

Live/next/done hero derived from the routine.

06 Teacher Features	Cancel/notice/undo + the cancellations table.
07 Resources	Semester folders, admin upload to Storage.
08 Notifications &	One insert -> Realtime feed + FCM push.
09 Profile	Read/update own row; the textbook layering example.
10 Admin & Engine	The differentiator: CP-SAT timetable pipeline.
11 Find Teacher	Schedule-derived real-time teacher locator.
12 Room Availability	Schedule-derived real-time room occupancy.

Say this: Each feature PDF ends with a 'Likely defense questions' table - rehearse those out loud. The answers are written the way you'd say them to the examiner.